

# Where the Logic Landed and What Was Proved: A Measured Die Map and a Formal Programme for a Neuromorphic System-on-Chip in an Open 130 nm PDK

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## Abstract

A companion paper [1] describes a RISC-V management core and an event-driven spiking accelerator hardened together on one die in the IHP SG13G2 open 130 nm process, and reports what the fault tolerance costs. This paper adds the two things it could not carry: *where* the design landed on the die, counted from the sign-off placement, and *what was proved* about it, as a programme rather than a remark.

The die map divides the  $4.79\text{ mm}^2$  die into the five regions the floorplan creates and counts what occupies each. Of 60,694 non-fill placed instances, 50,161 are in the central channel between the two macro rows; the core and the accelerator hold 21,526 and 13,638 of those. The peripherals reach out of the channel: the timer, the GPIO port and the QSPI controller are the three largest occupants of the ROM column’s lower pocket, the placeable area nearest the 48 of 94 top-level pins below the core box and the 39 to its right. Every block figure here is a *connectivity attribution* with a stated error bar, because only 1,140 of 168,592 instance names survive synthesis with a dot in them and exactly one carries a top-level block name; the residual is 9.747%, reported per region, where it runs from 0.93% to 50.24%.

The formal programme is 86 SymbiYosys tasks across 20 property sets, summed from the job files’ [tasks] sections: 79 PASS, two TIMEOUT, one UNKNOWN, four killed by hand. The event engine’s one-line invariant is proved over every reachable state in 5 s, where two documents had declined to prove it. The register file with its storage and scrub does not close with the real codec—four tasks under three solver back ends return no verdict, the two under yices stuck at step 4 on its parity network—and closes unbounded in 44 s once the codec is abstracted. riscv-formal leaves all eight M-extension instructions without a formal result. No silicon exists, and no frequency is published: wire is 1.9% of a violating path at the median, over 78 gate stages.

## 1 Introduction

The part is a single-die system-on-chip: a 32-bit RISC-V management core, 72 KiB of on-chip memory in six vendor SRAM macros with two more for check bits, a small set of peripherals, and one node of an event-driven spiking accelerator, hardened end to end through an entirely open flow on IHP’s open SG13G2 130 nm kit. Its architecture and its fault-tolerance scheme are the companion paper’s [1], compressed here to what this paper’s two contributions need.

**No silicon exists.** Nothing here has been fabricated. What has been submitted to a multi-project shuttle is a frozen slice of the accelerator alone: the TTIHP26b tile `tt_um_melihakbulut_nssoc`,  $6\times 2$  tiles at  $1,289.28\times 313.74\text{ }\mu\text{m}$ , pinned artefact by artefact by blob hash in the project’s

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freeze record, with a shuttle deadline of 2026-09-21. The system-on-chip instantiates that frozen slice out of its own directory rather than copying it, and is itself unfabricated.

**The two contributions.** **Where the design landed:** Figure 2 divides the die into the five regions the floorplan creates and annotates each with the blocks that actually occupy it, and §5 states both what it shows and the limits of the attribution behind it. And **the formal programme:** 20 property sets and 86 tasks, with block, property, task count and result, in Table 2, where the companion paper gives two sentences.

**Position.** The commercial reference is Frontgrade Gaisler’s GR801 [7], ahead of this design on process node, qualification and the fact that it will exist as silicon. This is not a competitor to it but an open one: every source, harness, measurement record and negative result is published, so each claim below can be re-derived rather than believed. An SNN on an open PDK is not itself a contribution and is not claimed as one [9, 8]; beam data [11, 6], the space survey [10] and the author’s own adjacent chip [5] are the companion paper’s related work.

## 2 Architecture, compressed

**Core and fabric.** lowRISC’s Ibex in its `small-pmp` configuration: RV32IMC, two-stage pipeline, flip-flop register file, four physical-memory-protection regions, and `SecureIbex` 0. It is vendored pristine at a pinned commit with no patch to Ibex itself; one module is substituted at its boundary—the architectural register file, for §3’s protected version—and that substitution is what §4’s riscv-formal runs compare against the stock core. The fabric speaks Ibex’s own request/grant/response protocol with two masters, six slave ports, round-robin arbitration and two outstanding requests; the peripheral bus is real AMBA 3 APB; and every unmapped address raises a bus error the core takes as an access fault.

**Memories and macros.** All eight macros are `RM_IHPSG13` 1P parts with byte masks: four  $2,048 \times 64$  for the system RAM, two  $1,024 \times 32$  for the 8 KiB boot ROM, two  $512 \times 16$  for the ROM’s check bits.  $2,337,520 \mu\text{m}^2$  of instance area, 48.8% of the die; Figure 1 gives sizes, origins and orientations. Every `RM_IHPSG13` part puts *all* of its signal pins on one edge, so the lower row is placed mirrored to face both rows’ pins into the channel between them—the one constraint that creates the five regions of §5.

**Peripherals and accelerator.** The blocks the die map labels are a core-local interruptor carrying a 64-bit free-running `mtime`, a general-purpose timer, a three-stage escalating watchdog in its own power-on reset domain, a console UART, a 16-pin GPIO port, a QSPI flash controller, a boot controller, a memory scrub engine and a bus and upset telemetry block; seven of the sixteen peripheral slots are reserved addresses with nothing behind them. The accelerator is one node of a sixteen-node address space—8 neurons, 8 axons, 64 four-bit signed synapses, a 72/64 Hsiao SECDED code on the weight word and 55 configuration bits triplicated in the netlist—and **it is driven exactly as external silicon would be:** one 40-bit mode-0 serial frame per register transaction, at 176 clock cycles per register access.

## 3 Hardening, compressed, and what it costs

The scheme is layered rather than uniform: a shortened SECDED code with a background scrub on the register file; four byte-wide 16/8 codewords in every 64-bit RAM row, so a byte store needs no read-modify-write; a 39/32 word code in two check macros on the boot ROM; triplication with a voter on the watchdog’s persistent state; `mtime` as the data field of a 72/64

| mechanism                              | measured cost                                                                                                      | measured against                                                                                                                                                     |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| register file SECDED and scrub         | +38,050.5762 $\mu\text{m}^2$ , +13.80 %, +222 flip-flops on the core                                               | a re-measured <code>small-pmp</code> baseline of 275,682.6198 $\mu\text{m}^2$ ; the core as shipped, with the later fast-correction path and fault port, is +14.64 % |
| system-memory protection               | +2,068 cells, +234 flip-flops, +31,391.3124 $\mu\text{m}^2$ (+4.783 %); the usable RAM halved to 32 KiB            | the whole SoC at the SRAM boundary; +3.01 % of placed standard-cell area after layout                                                                                |
| boot ROM check macros                  | +90,618.62 $\mu\text{m}^2$ (+4.03 %), two 512x16 parts                                                             | the six-macro area, costed from the vendor LEF, and now placed                                                                                                       |
| watchdog state triplication            | 715 cells and 102 flip-flops against 395 and 53; +5,091.0552 $\mu\text{m}^2$                                       | the same file at <code>HARDEN = 0</code> ; +88.9 % of the block, +1.85 % of the core                                                                                 |
| <code>mtime</code> as a 72/64 codeword | +7,026.8310 $\mu\text{m}^2$ , +8 flip-flops on the CLINT; +1,917.4050 $\mu\text{m}^2$ , +18 on the telemetry block | +42.04 % of the block, +2.549 % of the core                                                                                                                          |
| clock-gate enable split by arrival     | +1 flip-flop, +498 cells, +829.4454 $\mu\text{m}^2$ (+0.12 %)                                                      | the same design without the split, at synthesis                                                                                                                      |

Table 1: The protection costs, as measured pairs. Every row is the difference of two synthesised designs or two layouts made in one session against a reproduced baseline, not a block figure carried across a synthesis boundary. The percentages are arithmetic on the measured values. Its one post-layout figure, +3.01 % of placed cell area, is a six-macro pair (`s67ecc` against `s67base2`), not the sign-off layout.

codeword; and the accelerator’s configuration bits triplicated in the netlist. Every correction and every uncorrectable syndrome is counted in a telemetry block that lives in the power-on domain and survives the reset the watchdog causes.

One detail matters to §4: the register file’s code is the 40/32 shortening of this project’s one SECDED code, whose eighth check bit is identically zero over a 32-bit word—so the stored word is 39 bits and costs seven check flip-flops per register, not eight. That is not an observation but what `regfile_secDED.sby` proves, and it is why two documents here can call the same codec 40/32 and 39/32 without disagreeing.

One cost is not in Table 1’s area column: the register file’s correction sits *in series* with what it protects, since the decoder is on the register read path and that path feeds the ALU. That is why the worst timing endpoint in the sign-off layout is one of this mechanism’s own check bits (§6). What the protection does not cost is time: the whole-SoC cycle invariant is unmoved at 217,682 cycles with the RAM halved and the codec on every read.

## 4 The formal programme

### 4.1 What it is, and how it was counted

Table 2 is the whole of `hw/soc/formal`: twenty SymbiYosys job files and 86 tasks. **The count is the author’s own**—the twenty `[tasks]` sections were read and summed, and each task’s verdict then read from its own directory, a step that is load-bearing because `sby` writes a `status` file when it finishes and four tasks here have none. The project’s verification log last recorded this directory on 2026-09-11 at 64 tasks, all passing; it has grown by 22 since and the verdicts are no longer uniform, so that entry is superseded on this point and is named rather than quietly replaced.

| property set      | block             | what it states                                                                                                                                                                                                                           | tasks | result                          |
|-------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|---------------------------------|
| byte_secdec       | memory codec      | the 16/8 byte shortening corrects one error, detects two and never miscorrects, check-bit flips included                                                                                                                                 | 4     | 4 PASS                          |
| clkgate_wake      | clock-gate enable | T1, T2: the block is clocked by the end of the cycle after any frozen term rises; plus the fault case with the side assumption dropped                                                                                                   | 6     | 6 PASS                          |
| clkgate_wake_gnt  | clock-gate enable | G1–G3, T2: the same for a wakefulness-qualified grant, as a separate abstraction                                                                                                                                                         | 3     | 3 PASS                          |
| regfile_scrub     | register file     | R1–R4 on the real module at <code>SCRUB = 1</code> : a written word reads back while the scrub walks; the pointer visits <code>x1–x31</code> and never <code>x0</code> ; scrub and core never contend; a fault-free file raises no error | 5     | 1 PASS<br>2 TIMEOUT<br>2 KILLED |
| regfile_scrub_abs | register file     | the same four, with the codec abstracted to a stand-in satisfying $\text{dec}(\text{enc}(x)) = x$                                                                                                                                        | 5     | 2 PASS<br>1 UNKNOWN<br>2 KILLED |
| regfile_secdec    | register file     | the 40/32 shortening corrects one and detects two, and its eighth check bit is identically zero                                                                                                                                          | 4     | 4 PASS                          |
| soc_apb_bridge    | fabric            | A1–A11: AMBA 3 APB conformance of the bridge, as clauses of the specification                                                                                                                                                            | 3     | 3 PASS                          |
| soc_apb_wb        | fabric            | I1, W1–W8, P1–P3, D1–D2: APB completer to Wishbone B3 classic initiator                                                                                                                                                                  | 3     | 3 PASS                          |
| soc_boot          | boot controller   | D1–D10 at three parameter points including <code>HARDEN = 0</code> , so the unhardened baseline is proved to run identical policy                                                                                                        | 5     | 5 PASS                          |
| soc_bus           | fabric            | F1–F10 including no-starvation, the outstanding limit, decode against the generated map and the clock-gate enable; run again at <code>REQ_REG = 1</code>                                                                                 | 6     | 6 PASS                          |
| soc_busstat       | telemetry         | B1–B6: saturation, stickiness and the interrupt, narrow and at the shipped width                                                                                                                                                         | 4     | 4 PASS                          |
| soc_clint         | CLINT             | C1–C4, P1–P3 from the privileged specification and the slave contract; E1–E2: the stored <code>mtime</code> is always a codeword and the codec is invisible when it is                                                                   | 3     | 3 PASS                          |
| soc_gpio          | GPIO              | P1–P7 reconstructed from the ports and the manual, at 4 and at the instantiated 16 bits                                                                                                                                                  | 4     | 4 PASS                          |
| soc_mem_ecc       | memory codec      | M1–M6 over a four-row array: every row is a codeword in every reachable state, the codec is invisible, the scrubber never takes the port from the bus                                                                                    | 3     | 3 PASS                          |
| soc_npu           | event engine      | H5: the strobe presenting an event to the node is exactly the <code>E_PIN_S</code> state—equal, not implied—plus its reachability                                                                                                        | 3     | 3 PASS                          |
| soc_npu_ser       | serial transport  | P1–P2 frame format and H1–H4 host obligations of the frozen die’s own protocol; D1–D5 including the frame bound; two rate points                                                                                                         | 4     | 4 PASS                          |
| soc_qspi          | QSPI              | Q1–Q8 reconstructed from the APB port and the pins, narrow and at the shipped widths                                                                                                                                                     | 4     | 4 PASS                          |
| soc_scrub         | scrub engine      | C1–C7 on the counters and the scrubbers’ control, narrow and at the shipped counter width                                                                                                                                                | 4     | 4 PASS                          |
| soc_wdog          | watchdog          | W1, W3–W5 of the block’s own requirements, plus W7–W9 for the window, the kick budget and the registered reset request, and the set again with the window disabled                                                                       | 4     | 4 PASS                          |
| soc_wdog_tmr      | watchdog          | T1–T6: three replicas under one voter round-trip their storage transforms, agree without a fault, mask any fault confined to one replica and report it exactly; five widths, two engine families                                         | 9     | 9 PASS                          |
| 20 sets           |                   |                                                                                                                                                                                                                                          | 86    | 79 PASS                         |

Table 2: Every property set under `hw/soc/formal`. The task counts are the entries in each job file’s own `[tasks]` section, summed by the author over the twenty files rather than taken from any document; the verdicts are each task directory’s `status` file or, where sby wrote none, the last line of its `logfile.txt`. The frozen accelerator slice carries eight further property sets and

Of the 86: 79 PASS, two TIMEOUT at a bound the job file sets, one UNKNOWN, four with no verdict because they were stopped by hand. **Those are four different results and the table uses the word the tool used.** A TIMEOUT reached a bound its author wrote down; a killed job is the absence of a result, no bound having been set before a person sent a signal; an UNKNOWN is sby’s own word for a  $k$ -induction run whose base case passed and whose induction step did not close at that  $k$ . None is a FAIL, and no property here has produced a counterexample from reset.

## 4.2 The event engine: one line, over every reachable state

`soc_npu.v` is the largest file in this project’s own RTL and the block the accelerator lives in, and until this job it carried no proof. Two documents named the gap and declined it; the second wrote the sentence this job answers—that the invariant `aer_in_stb == (ev_state == E_PIN_S)` “is one line and is the entire basis of the mechanism, and it is checked here by one cocotb test and one textual guard rather than over all reachable states.” The property says the strobe presenting an event to the node *is* that one FSM state: not implied by it, not implying it—equal, so an upset that moves the state moves the strobe with it and an upset that flips the strobe alone is a mismatch the equality exposes. **It is now proved by  $k$ -induction over every reachable state in 5 s of clock time**, with a bounded check to depth 30 and a cover task showing the state reachable so the assertion is not vacuous.

One thing makes it a smaller claim than it looks, and the job file says so: the invariant has *no flip-flops of its own*—its mechanism is a comparator and an AND—so a mapper that folded the gate away would leave every flip-flop census in this project unchanged, which is why its netlist guard is a mutation test requiring a strictly smaller netlist at the same flip-flop count. Liveness is not proved either.

## 4.3 The register file: where the codec stops the solver

The job `regfile_scrub.sby` binds the design’s own register file at `SCRUB = 1` with nothing abstracted, under a wrapper stating the four properties Table 2 lists. The cover task passes, reaching all thirty-one visits at depth 40. **Nothing else does.** The  $k$ -induction and bounded tasks under `yices` sat at step 4 and were stopped by hand after twelve hours—KILLED, because no bound had been set on them—and two further tasks added with bounds written into the job file, `abc_pdr` and `smtbmc boolector`, both report DONE (TIMEOUT, `rc=8`) at 10,800 s. Four tasks, three solver back ends, no verdict.

Step 4 is the first cycle at which the read-back property examines a stored word: the solver is asked to relate eight parity trees at the write encoder to eight at the read decoder through thirty-one scrub encoders on the same storage. Parity is the classic case conflict-driven SAT cannot do in polynomial resolution, and none of the pinned solvers carries Gaussian elimination. **The cost is structural, and two sources disagree about its shape:** the stand-in codec’s header says all four engines stall at step 4, and the boolector run in this tree does not—it cleared step 4 in 13 minutes, step 5 in 30, step 6 in 59, and was still inside step 9 three hours in.

`regfile_scrub_abs.sby` is the same job with the codec replaced by a stand-in with the same module names and ports and a degenerate code. Under it `abc_pdr` proves all four properties for every reachable state in 44 s. The composition is the usual one: a property of the wrapper that holds for every codec satisfying  $\text{dec}(\text{enc}(x)) = x$  holds for the real codec, whose  $\text{dec}(\text{enc}(x)) = x$  is `regfile_secded.sby`’s separate theorem, and what it does *not* cover is any property depending on the code’s distance. That job’s own bounded tasks reached step 23 after 43 h 30 min and were then stopped by hand (KILLED again), which prices the bounded route at 43 hours of two cores for an answer the unbounded one had in 44 s. The one UNKNOWN is a  $k$ -induction task at  $k = 12$  on the same job whose base case passed and whose induction step failed on the error-flag property: a statement about  $k$ , not about the design.

#### 4.4 riscv-formal, and the eight instructions with no result

The management core was run against riscv-formal in two work trees: the stock Ibex, and the same core with the protected register file substituted. On the stock core 70 of 79 generated bounded checks pass, and all 79 cover jobs pass on both trees; every consistency check except liveness passes, both program-counter checks, causality and uniqueness among them. On the substituted core—the one this design ships—the count is 69, and the check that moved is the register file’s own: `reg_ch0` reports `DONE (ERROR, rc=16)` after a signal from outside the run at 1 h 41 min with no bound set on it. **An ERROR is not a TIMEOUT**, and this is one reason the register file has a job of its own.

**What does not pass is the whole M extension, and in no case is the reason the core.** Divide and remainder fail because riscv-formal’s own models compute an *unsigned* division and remainder—a diagnosis resting on two counterexamples checkable by hand, on IEEE 1364-2005 section 5.5.1, and on an independent simulator agreeing with Yosys. Re-running the two against a corrected model made the proof *harder*: they had proved nothing after 41 and 49 minutes, where against the defective model they had failed in 852 and 731 seconds. The remaining six—`mul`, `mulh`, `mulhsu`, `mulhu`, `divu`, `remu`—were stopped without a verdict after 2,117 to 2,419 seconds each, then run again on both cores under a second solver with `timeout 7200` written into each generated job file *so that the recorded status would be the tool’s own word*: **all twelve report TIMEOUT and none closes**. Yet every one of the six passes its cover job, in 9 to 73 seconds under the first solver and 1,388 to 10,999 under the second. The witness is easy and the proof is not, so the obstacle is the bit-blasting solver and not the model or the depth—and the evidence for the M extension here remains one simulated case per operation.

#### 4.5 What formal bought, and what it did not reach

It bought three things simulation did not. **Conformance as a theorem**: the APB bridge’s compliance with AMBA 3 and the fabric’s no-starvation are proved rather than exercised, which is the stated reason APB was implemented and AHB was not. **Masking as a theorem**: the watchdog’s triplicated word is proved to mask any fault confined to one replica and report the disagreement exactly, at five widths and two engine families, and the boot controller’s identical composition relies on that theorem instead of rebuilding it. And **invariants over all reachable states** for what a test can only sample: every stored memory row is a codeword, the stored `mtime` is always a codeword, the event strobe is exactly one FSM state.

It did not reach four things, each named in the job files themselves. **No fault model**—not one job here injects a fault, because a formal harness that deposits into storage is proving a property of the harness, so every correction claim in this design is a seeded fault-injection measurement. **No liveness**, except where a cover shows a behaviour reachable. **Not the accelerator’s clock gate on the real block**—`soc_npu.v` instantiates the frozen die, so `clkgate_wake.v` is an abstraction and not a block of this SoC. And **not the anti-merge argument**: the transforms that stop the synthesiser hashing three replicas into one are properties of the mapped netlist, carried by a census and not by a proof.

### 5 Physical implementation and the die map

#### 5.1 The flow, and which run every number is from

The flow is entirely open: Yosys for synthesis, OpenROAD through LibreLane for place and route, OpenSTA for timing, Magic and KLayout for design-rule checking, Netgen for layout-versus-schematic, KLayout again for XOR, on IHP’s open SG13G2 at a pinned commit. **Every physical figure in this paper is from one run, s83romecc5**—the eight-macro sign-off

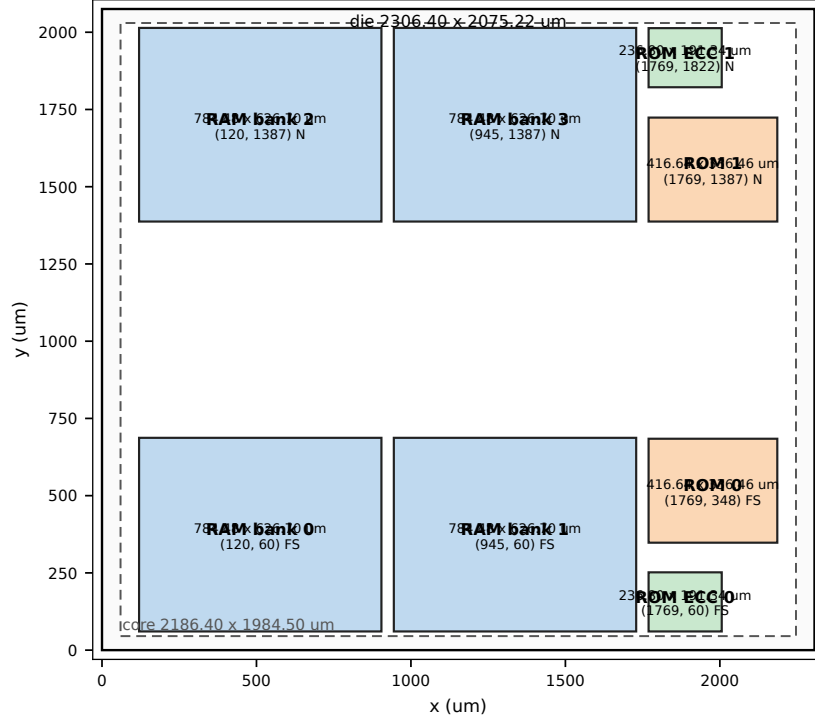


Figure 1: The eight vendor macros with sizes, origins and orientations on the  $2,306.40 \times 2,075.22 \mu\text{m}$  die; no standard cells drawn. The lower row is mirrored so that both rows’ pin edges face the channel between them.

layout—unless the sentence quoting it names another. That matters because this corpus has repeatedly measured one layout and quoted it beside another: the companion paper’s own physical tables come from a six-macro run containing no accelerator at all. The flow is bit-reproducible here, so a difference between two layouts is a difference of netlists and never a draw.

## 5.2 The floorplan and its five regions

The die is  $2,306.40 \times 2,075.22 \mu\text{m} = 4,786,290 \mu\text{m}^2 = 4.79 \text{ mm}^2$ , around a site-aligned core box of  $4,338,910 \mu\text{m}^2$ . The eight macros are FIXED; nothing else was told where to go. Two mirrored macro rows and the ROM column between them and the die edge partition the placeable area into exactly five regions: the **central channel**,  $700.08 \mu\text{m}$  tall and the full core width; the **lower** and **upper macro rows**, the slivers left beside and between the RAM macros; and the **lower** and **upper halves of the ROM column**, the pockets the ROM macros—shorter than the RAM—leave at the right of each band. Every non-fill placed instance falls in one of the five, and they sum to 60,694, the whole non-fill population.

## 5.3 The attribution, and the error bar on every block figure

**This must be stated before the map is read.** “Where is the processor” is a question about a die carrying 168,592 instances, and after this flow’s synthesis almost none of them says which block it came from. Only 1,140 of those names contain a dot at all—632 the RAM’s and 266 the ROM’s, tie cells but for the eight macros themselves, most of the rest clock buffers—and by the attribution script’s own prefix test **exactly one instance in the die carries a top-level block name**: `u_ibex.core_clock_gate_i.u_icg`, an integrated clock gate. Everything else is a Yosys name with no block in it, so a picture colouring only the named instances would colour one cell.

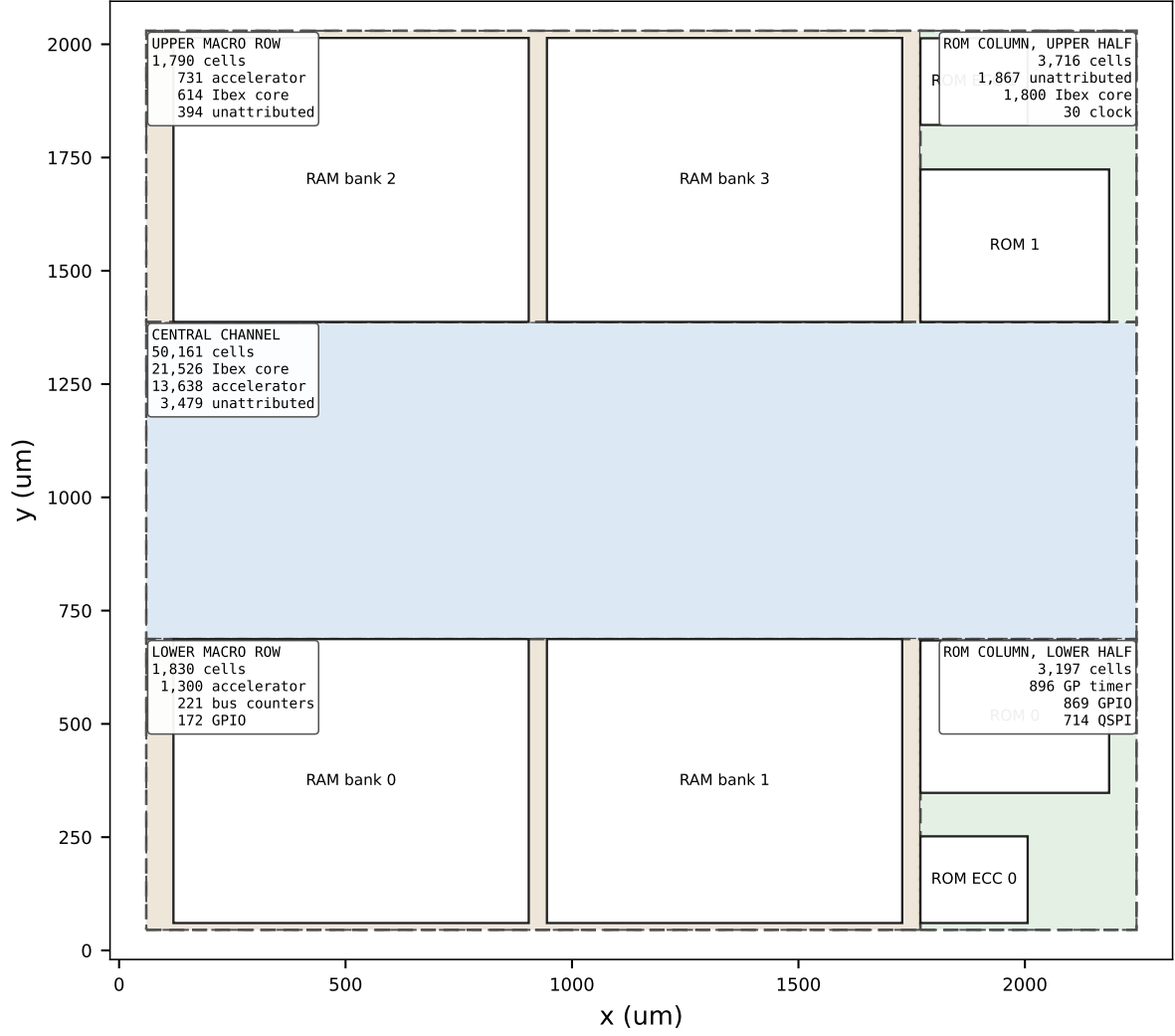


Figure 2: **The die by region.** The five regions the floorplan creates, each annotated with its non-fill cell count and its three largest occupants, over the placed cells and the eight fixed macros. Occupancy is counted from the sign-off placement, not assigned by intent: apart from the macros, nothing here was constrained. The boxes rank the unattributed and clock-tree classes among the occupants where they fall, which is why three of the five name the unattributed one and one of those three also names the clock; Table 3 separates the unattributed share out.

| region                 | cells  | three largest attributed occupants                  | unattributed    |
|------------------------|--------|-----------------------------------------------------|-----------------|
| central channel        | 50,161 | Ibex core 21,526; accelerator 13,638; CLINT 1,739   | 3,479 (6.94 %)  |
| lower macro row        | 1,830  | accelerator 1,300; bus fault counters 221; GPIO 172 | 17 (0.93 %)     |
| upper macro row        | 1,790  | accelerator 731; Ibex core 614; RAM control 4       | 394 (22.01 %)   |
| ROM column, lower half | 3,197  | GP timer 896; GPIO 869; QSPI 714                    | 159 (4.97 %)    |
| ROM column, upper half | 3,716  | Ibex core 1,800; ROM control 18; accelerator 1      | 1,867 (50.24 %) |
| total                  | 60,694 |                                                     | 5,916 (9.747 %) |

Table 3: What each region holds, counted from `s83romecc5`’s final DEF and netlist. “Cells” is non-fill placed instances: fill, decap, tie, tap and antenna cells are excluded. The clock tree is not: its 1,075 cells belong to no block, so they are inside the “cells” column and outside the occupant columns. The unattributed share is this attribution’s own error bar, and it is not uniform. This project also runs a *different* five-zone census over the same DEF, which counts 64,349 standard cells because it excludes only fill; the two partitions answer different questions and their rows must not be compared.

The attribution is therefore by *connectivity*. Yosys renames cells but keeps the hierarchical name of any net it did not create, so 643 nets still vote for a top-level block; each anonymous cell takes the block most of its pins’ nets vote for; unattributed cells then take the majority block of their neighbours, twice, which is where it converges. Clock buffers, tie, tap, fill and decap cells are recognised by name or master and form their own classes, belonging to no block.

**The residual is reported, not hidden.** 5,916 of the 60,694 non-fill instances—9.747 %—end unattributed and are drawn grey. That is the error bar on every block-level statement here and on every block figure this project has published, Figure 3 included. It is also not uniform, which nothing has stated before: Table 3 gives it per region, from 0.93 % in the lower macro row to 50.24 % in the upper half of the ROM column. A block count from that last region is worth little; one from the channel, at 6.94 %, is worth a good deal more.

## 5.4 What is in each region, and why

**The channel holds the core and the accelerator, because it is the only contiguous area.** 50,161 of 60,694 non-fill instances—82.6 %—are in the channel, and the two largest blocks dominate it: 21,526 of the core’s 23,942 attributed cells and 13,638 of the accelerator’s 15,686. The placer was not told to do that; the channel is simply the one region tall and wide enough to take a block of tens of thousands of cells without splitting it. The overflow goes into the macro rows, where the accelerator is the largest occupant of both: 1,300 cells in the lower row and 731 in the upper.

**The CLINT’s closure is entirely in the channel, which settles an open question.** A separate census of its 1,802-cell register-to-register closure puts 1,802 of 1,802 in the channel, none in either pocket and none in the strip beside the ROMs. A prior document had ended with the instruction that keeping that set on one side of the ROM “is a floorplan requirement and not a placement one”. **It is already met, by the placer, with nothing asked of it**—and five earlier runs that forced the cluster into a box or moved a fence were answering a question the eight-macro floorplan had already answered by accident.

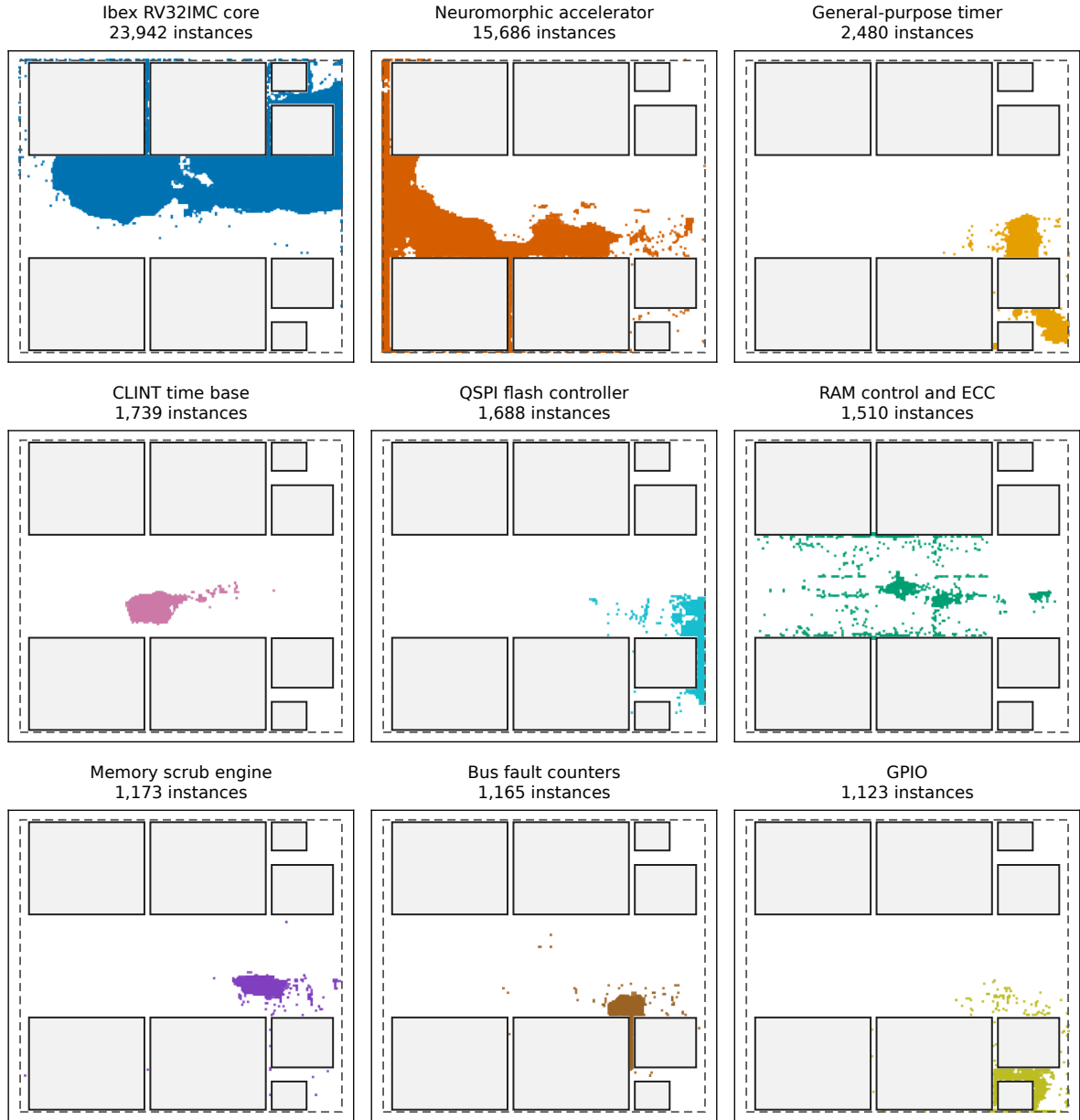


Figure 3: One panel per block, for the nine largest blocks by attributed cell count—fourteen clear the script’s 300-cell floor and it keeps the nine biggest—over a grey ground of every other non-fill instance. These carry the residual of 5: 9.747% of the non-fill instances are in neither the coloured set nor any other block’s.

| corner              | setup worst | setup violations | hold worst |
|---------------------|-------------|------------------|------------|
| nom_slow_1p08V_125C | -7.5758 ns  | 3,529            | +0.4261 ns |
| nom_typ_1p20V_25C   | +2.6396 ns  | 0                | +0.1769 ns |
| nom_fast_1p32V_m40C | +8.3630 ns  | 0                | +0.0296 ns |

Table 4: Post-layout static timing on s83romecc5 at the three corners, on extracted parasitics, with a 5 % derate applied as a float and 0.25 ns of clock uncertainty, against a 20 ns SDC period. Setup total negative slack at the slow corner is -14,735.65 ns; **hold has zero violations at all three corners**. The design also fails max slew (533/77/20 violations at slow, typical and fast), max capacitance (95/96/98) and max fanout (589 at every corner, in a flow with no max-fanout checker step at all). Three of those are visible only because the violation-corner keys were bound to every corner before the run rather than audited after it; under the shipped defaults the flow would have printed warnings and exited zero.

**The ROM column’s lower pocket is peripheral ground, because that is where the peripherals’ pins are.** Its three largest attributed occupants are the general-purpose timer (896), the GPIO port (869) and the QSPI controller (714), with the boot controller fourth at 463. GPIO and the boot controller have their plurality here—869 of 1,123 and 463 of 891—while the timer keeps 1,562 of 2,480 in the channel and QSPI 972 of 1,688: the pocket pulls the peripherals out, it does not hold them. That is not where a block diagram would put any of them and it is exactly where the I/O placement puts their pins. Of the 94 top-level pins in the DEF’s PINS section, **48 lie below the core box and 39 to its right**—87 of the 92 signal pins on the bottom and right edges—with four on the left, one at the top and the two power pins inside the core. GPIO alone accounts for 42 bottom-edge pins and seven right-edge ones; QSPI has sixteen, all on the right. **There is no pad ring:** these are DEF pin shapes 0.2  $\mu$ m in from the die boundary, placed by the flow’s I/O placement step, and the design has no pads, no package and no I/O cells.

And the upper half of the ROM column is where the attribution runs out: 1,867 of its 3,716 cells are unattributed and 1,800 attributed to Ibex, which is as much as this method can say.

## 5.5 Sign-off

Three things travel with Tables 4 and 5.

**The layout does not meet its setup constraint**, reported with the same weight as what passes: -7.5758 ns at the slow corner on 3,529 violating endpoints against a 20 ns period. The aggregate metric `design__violations` nonetheless reads 0 in this run’s own metrics file. **The aggregate metric does not aggregate**, and a reader who trusted it would conclude the layout was clean.

**It meets hold at all three corners, and that is not free.** Raising the antenna repair’s iterations and margin clears both antenna violations—s83ant reports 0 nets and 0 pins—for 60 additional diodes, and *loses hold*: -0.4351 ns on 55 endpoints, 38 fast, 16 typical, one slow. That is a bad trade, a hold violation being a functional failure where the two antenna violations it removes are ratio overshoots, and it cannot be tuned away: repeating the repair at gentler settings converges on exactly the same layout.

**The power figure is an estimate wearing a measurement’s clothes.** The flow calls `report_power` with nothing having read a VCD or a SAIF, so 56.41 mW rests on the timer’s default propagated activity and not on any program this project has run. Activity-annotated measurements on layouts containing the accelerator give 22.142 mW computing and 8.287/8.315 mW idle with the core gated in WFI, and those are the figures to quote.

| check                       | result on s83romecc5                                                                                                                                                                                                                  |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| detailed-route DRC          | 0, at the router’s iteration 16                                                                                                                                                                                                       |
| disconnected pins           | 0                                                                                                                                                                                                                                     |
| power-grid violations       | 0 on both rails                                                                                                                                                                                                                       |
| antenna, post-route         | 2 violating nets, 2 violating pins; 1,258 diodes inserted                                                                                                                                                                             |
| routed wirelength           | 5,746,921 $\mu\text{m}$ ; 62,767 signal nets and two special; 486,886 vias, all single-cut; longest net 4,292.89 $\mu\text{m}$                                                                                                        |
| Netgen LVS                  | circuits match uniquely, 263 symmetries, all seven difference counters 0, with the vendor macros black-boxed on both sides—so it signs off the connectivity <i>into</i> all eight macros’ pins and says nothing about their internals |
| KLayout DRC                 | 11,048 markers, 6,584 distinct shapes, and <b>every one inside the vendor macro hierarchy</b> ; 0 outside. The deck declares 173 rule categories in 47 groups; three fired                                                            |
| Magic DRC, abstracted       | 170 error boxes, 20 inside a macro footprint and 150 outside—and every one of the 150 within 0.500 $\mu\text{m}$ of a macro edge, none in open routing area                                                                           |
| KLayout XOR                 | <b>not run on this layout</b> ; run on the antenna-repaired variant s83ant, where the difference count is 0                                                                                                                           |
| total power, typical corner | 56.41 mW, <i>with no switching activity annotated</i>                                                                                                                                                                                 |

Table 5: Sign-off on the eight-macro layout. Route, pin, grid, antenna, wirelength and power are s83romecc5’s own metrics; the four deck rows are decks run on that layout in trees of their own, its configuration gating all four off—s83lvbbb2, s83kdrc, s83mdrc and, for the XOR, s83antxor. The two geometric deck rows need the qualifier the companion papers carry [4]: the vendor SRAM macro does not pass this PDK’s own decks, which is a property of the kit rather than of this design, so the honest form of a deck result here is two numbers—inside the macro and outside it—and not one.

## 6 What the measurements settle

### 6.1 No frequency, and now a reason as well as a rule

This project publishes no clock frequency for this design, and the rule is its own: a figure published externally must be traceable to measurement or clearly labelled as a design target, and the only figure available is a measured *non-closure* from a layout that fails setup, slew and capacitance. Two further grounds stand: the mission requirement is a frame rate, and the shuttle tile would carry a different figure, so publishing two numbers would tell a reader the project cannot say what its clock is. The 20 ns SDC period is kept as a flow constraint so that four rounds of slack measurements stay comparable. What §6.2 adds is the measured reason.

### 6.2 The timing is not a floorplan question

Every violating path in the sign-off report was decomposed at its launch flip-flop’s Q, each incremental delay after it attributed to a cell arc or to a net, with the clock network counted into neither so the percentages compare across layouts with different clock trees. Over the 1,000 violating paths: **wire is 1.9 % of a violating path at the median and 3.0 % at the most wire-bound path**, over a median of **78 gate stages**, 57 to 91. The worst path is 77 stages, 26.2904 ns of cell delay and 0.7827 ns of net delay—so if every wire on it were made ideal it would still take 26.29 ns against a 20 ns constraint. A floorplan moves wire; there is no version of this floorplan that moves 78 gate stages. Nor is the profile an artefact of this one: an earlier layout on a different macro set, s71boot, gives 1.9 % at the median and 2.8 % at the maximum over its 985 violating paths. That is the second half of the CLINT result above—the condition five placement experiments were run to enforce is already met, and the quantity they could

have moved is under 5 % of every violating path in any of the three floorplans measured.

**A correction belongs here.** These figures were first published on 2026-09-16 as 3.6 % at the median and 4.7 % at the maximum over a median of 88 stages, from a tool that was cutting each path at the end of the report block instead of at `data arrival time` and so counting the *capture* clock’s tree into the data path. It was corrected the same day, and the old figures are an instrument defect rather than a superseded measurement—nothing about the design changed between them. Three places in the corpus still carry a pre-correction figure: the summary section of the document that made the correction and the opening of the document that cites it carry the 88, and the document index carries the 3.6 % beside a corrected 3.0 %. All three are named here.

The register file’s data bits hold 1,007 of the 3,529 violating endpoints, its SECDDED check bits 217 and the rest of Ibex 864—2,088 together, 59 % of the count and 58 % of the total negative slack—against 54 for the fabric. **And the single worst endpoint is a SECDDED check bit of the register file:** read address, through the core’s datapath, into the write encoder, into the check-bit storage. It is the hardening’s own write path, priced in §3 in area and in fault coverage but never in depth. No placement removes an XOR tree.

### 6.3 The registered request phase buys the depth and does not route

The one structural change inside this project’s own RTL that the path decomposition points at is a registered request phase in the fabric: make the grant and the transfer two events instead of one, so the slave’s decode and read multiplexer start at a flip-flop. It was built behind a parameter, the default proved unchanged by an equivalence miter, and the fabric’s whole property set proved again at both settings—the six tasks in Table 2’s `soc_bus` row.

**It buys what it was built to buy.** Maximum combinational depth goes from 81 levels to 69 and the median from **39 to 20**; the deepest endpoint class, the CLINT’s response registers, goes from 81 to 16. **It costs 24.89 % of the cycle count:** the same workload runs in 520,398 cycles against 416,673, passing at both. The register is a skid buffer, so throughput is unchanged; what grows is latency, by one cycle, on every transaction.

**And its layout does not route.** The same netlist through the same flow with the same configuration and initial state, run `s89rr1`, stops at global routing with congestion: 387 overflowing tiles against 0 for the sign-off layout, 353 of them on one metal layer, with total wirelength up 586,339  $\mu\text{m}$  (+8.2 %) for 159 additional nets. Before the change, the fabric’s address, byte-enable and write-data outputs are combinational selections between two masters, driven from wherever the placer put the masters’ own logic; after it, all three come out of *one* register that must reach all seven targets—the six slave ports and the internal error slave—which, as §5 measured, the I/O placement and the macro column spread to opposite corners of the die. **A single point driving a 69-bit bus to seven destinations is a broadcast, and a broadcast goes on the long layer.** The change adds wire not because it adds cells but because it moves the fan-out’s origin from two well-placed sources to one.

So the trade on offer is not a quarter of the cycle count against an unmeasured slack improvement, but a quarter of the cycle count against a design that stops at global routing, and **the default stays off.** The depth result stands, being a netlist measurement independent of placement; what the layout adds is that *this* way of delivering it—one register at one point—is what the router cannot take. A version registering the request at each slave would buy the same depth without the broadcast; it is not built, and its first constraint is known before it is.

### 6.4 And what none of this is

Not a flight part and not radiation-hardened. No silicon, no beam, and the open kit publishes no total-ionising-dose or single-event data, so every reliability figure in this project is a seeded fault-injection measurement on a model—including the campaign that once produced bit-identical

output on a defective design and its repair, a companion paper’s subject [3]. The spacecraft interfaces, the accelerator mesh and any design-for-test are absent, and the vendor SRAM macro cannot be signed off on this PDK version at all, which is why Table 5’s deck rows are two numbers and not one. The companion paper [1] carries the full list.

## 7 Availability

The public repository is <https://github.com/Melihakbulut221/nssoc>.

It is a generated mirror of the development tree, carrying the tree and not the history: the RTL, the testbenches, the formal jobs and their property files, the flow configurations, the register-map and memory-map single sources, the golden models and their suite, the shuttle submission, the numbered document corpus, and the LaTeX sources of the thesis [2] and of every preprint tracked there, this one included at the mirror’s next generation, with the thesis also shipping as a built PDF. Run trees are not in it, which is why every physical figure here is quoted with the run tag it came from.

**Three things are held back and none is technical:** the product-line positioning section of one document, the funding and shuttle commercials of a second, and the drafted text of an unsubmitted funding application. Each is replaced by a stub keeping the filename and saying what was held and why, so the documents citing them still resolve and a reader sees a withholding rather than a silent gap. **No negative result, no withdrawn claim, no measurement and no failing check is held.** That is the condition under which the rest of the corpus means anything, and it is why this paper can report a layout that does not close, a remedy that does not route, four formal tasks killed by hand and eight instructions with no formal result.

## 8 Conclusion

**Where it landed.** The channel holds the core and the accelerator because it is the only contiguous area; the ROM column’s lower pocket is peripheral ground because that is where 87 of 92 signal pins are; the CLINT’s closure is entirely in the channel, which retires five placement experiments. Each is a connectivity attribution carrying a 9.747% unattributed remainder, reported per region because it is not uniform: a die map of a flat netlist is an inference, and this one states its own error bar.

**What was proved.** 86 tasks across 20 property sets, counted from the job files: conformance and masking as theorems, invariants over all reachable states for what a test can only sample, one event-engine invariant closed in 5s that two documents had declined to close, one register-file job that does not close because parity is where the solver stops, and eight M-extension instructions with no formal result. And two negative results worth more than either: the timing is not a floorplan question, and the one structural remedy this project’s own RTL can reach halves the median logic depth, costs a quarter of the cycle count, and does not route. Both are published, because a record carrying only what worked is not a record.

## References

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